

# Starship HLS-Inspired Lunar Descent

Technical Handoff: Methods, Equations, and Propellant Budget

*Reference simulation: V17; optional site-access context: V18*

**Purpose.** This document provides a report-ready description of the modeled orbit-to-surface descent, the numerical methods and assumptions, final calculated quantities, validation evidence, and recommended MATLAB files/figures. All vehicle performance inputs are illustrative assumptions, not verified Starship HLS specifications.

## 1 Objective and Scope

Estimate propulsive velocity increment ( $\Delta v$ ) and propellant consumption for a simulated landing at latitude  $-88.811008^\circ$ , longitude  $123.690068^\circ$ , starting in a 100 km circular lunar orbit. The model comprises an impulsive deorbit maneuver, a ballistic two-body coast, and a finite-thrust guided descent. A discrete search minimizes powered-descent propellant subject to touchdown-speed, landing-error, and final-mass constraints. Surface terrain and landing-site suitability are handled separately.

## 2 Consolidated Results

| Quantity                                              | V17 result / derived estimate                        |
|-------------------------------------------------------|------------------------------------------------------|
| Initial mass before deorbit                           | 250,000 kg                                           |
| Deorbit impulse                                       | $19.455 \text{ m s}^{-1}$                            |
| Deorbit propellant                                    | $\approx 1,413 \text{ kg}$                           |
| Mass after deorbit                                    | $\approx 248,587 \text{ kg}$                         |
| Powered-descent propellant                            | 110,850 kg                                           |
| Powered-descent propulsive $\Delta v$                 | $\approx 2,026.6 \text{ m s}^{-1}$                   |
| <b>Total modeled propulsive <math>\Delta v</math></b> | <b><math>\approx 2,046.1 \text{ m s}^{-1}</math></b> |
| <b>Total modeled propellant</b>                       | <b><math>\approx 112,263 \text{ kg}</math></b>       |
| Touchdown mass                                        | $\approx 137,737 \text{ kg}$                         |
| Mass above assumed dry mass                           | $\approx 17,737 \text{ kg}$                          |
| Selected coast fraction / guidance time               | 0.928 / 660 s                                        |
| Powered-descent duration                              | 665.02 s                                             |
| Touchdown speed                                       | $1.8028 \text{ m s}^{-1}$                            |
| Landing-site error                                    | 3.6949 m                                             |

**Interpretation:**  $\Delta v$  is the accumulated *propulsive* increment,  $\int T/m dt$ , not the magnitude of the difference between the initial and final velocity vectors. Totals cover deorbit and powered descent only; they exclude Earth–Moon transfer, lunar orbit insertion, ascent, and unmodeled operational reserves. The rounded powered-descent  $\Delta v$  is derived from the rocket equation and reported mass consumption.

### 3 Assumed Physical and Vehicle Parameters

| Parameter                                | Assumed value                                           |
|------------------------------------------|---------------------------------------------------------|
| Lunar gravitational parameter, $\mu$     | $4.902800066 \times 10^{12} \text{ m}^3 \text{ s}^{-2}$ |
| Spherical lunar radius, $R$              | 1,737,400 m                                             |
| Initial circular orbit altitude, $h_o$   | 100,000 m                                               |
| Transfer perilune altitude, $h_p$        | 15,000 m                                                |
| Initial spacecraft mass, $m_0$           | 250,000 kg                                              |
| Dry mass, $m_{\text{dry}}$               | 120,000 kg                                              |
| Maximum thrust, $T_{\text{max}}$         | $2.0 \times 10^6 \text{ N}$                             |
| Specific impulse, $I_{\text{sp}}$        | 350 s                                                   |
| Standard gravity, $g_0$                  | $9.80665 \text{ m s}^{-2}$                              |
| Minimum commanded ON throttle            | 10% of $T_{\text{max}}$                                 |
| Maximum thrust-magnitude rate            | $20,000 \text{ N s}^{-1}$                               |
| Illustrative thrust-direction slew limit | $2^\circ \text{ s}^{-1}$                                |
| Touchdown-speed limit / preferred target | $2.0 / 1.85 \text{ m s}^{-1}$                           |
| Maximum landing-site error               | 100 m                                                   |
| Minimum final mass                       | 125,000 kg                                              |

### 4 Governing Equations and Method

#### 4.1 Impulsive deorbit maneuver

A retrograde impulse transitions the spacecraft from a 100 km circular orbit to an elliptical transfer orbit with a 15 km perilune. With apolune and perilune radii,

$$r_a = R + h_o, \quad r_p = R + h_p, \quad a = \frac{r_a + r_p}{2}, \quad (1)$$

$$v_c = \sqrt{\frac{\mu}{r_a}}, \quad v_a = \sqrt{\mu \left( \frac{2}{r_a} - \frac{1}{a} \right)}, \quad (2)$$

$$\Delta v_{\text{deorbit}} = v_c - v_a \approx 19.455 \text{ m s}^{-1}. \quad (3)$$

The ideal rocket equation gives the mass remaining after the impulse:

$$m_{\text{start}} = m_0 \exp\left(-\frac{\Delta v_{\text{deorbit}}}{I_{\text{sp}} g_0}\right), \quad (4)$$

$$m_{p,\text{deorbit}} = m_0 - m_{\text{start}} \approx 1,413 \text{ kg}. \quad (5)$$

This treatment assumes an instantaneous, ideal retrograde maneuver rather than a finite-duration deorbit burn.

#### 4.2 Ballistic coast

The transfer coast is propagated using a central, spherical lunar gravitational field:

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r}. \quad (6)$$

The half-period of the transfer ellipse is

$$t_{1/2} = \pi \sqrt{\frac{a^3}{\mu}}. \quad (7)$$

MATLAB's `ode45` integrates the coast, and powered descent begins at a chosen fraction of this half-period. The selected fraction was 0.928.

### 4.3 Finite-thrust powered descent

The three-dimensional equations of motion and mass depletion are

$$\dot{\mathbf{r}} = \mathbf{v}, \quad (8)$$

$$\dot{\mathbf{v}} = -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r} + \frac{T_{\text{act}}}{m} \mathbf{q}, \quad (9)$$

$$\dot{m} = -\frac{T_{\text{act}}}{I_{\text{sp}} g_0}, \quad \|\mathbf{q}\| = 1, \quad (10)$$

where  $\mathbf{q}$  is the actual thrust direction. The accumulated powered-descent propulsive increment is

$$\Delta v_{\text{powered}} = \int_0^{t_f} \frac{T_{\text{act}}(t)}{m(t)} dt = I_{\text{sp}} g_0 \ln \left( \frac{m_{\text{start}}}{m_f} \right). \quad (11)$$

The second equality assumes constant specific impulse, as in the model.

### 4.4 Feedback guidance and actuator approximations

The illustrative feedback controller commands an acceleration based on target position and current velocity:

$$\mathbf{a}_{\text{req}} = -\frac{6(\mathbf{r} - \mathbf{r}_{\text{target}})}{\tau^2} - \frac{4\mathbf{v}}{\tau} - \mathbf{g}, \quad (12)$$

$$\tau = \max(t_{\text{go}} - t, 35 \text{ s}), \quad \mathbf{g} = -\frac{\mu}{\|\mathbf{r}\|^3} \mathbf{r}. \quad (13)$$

Near the surface, inward commanded radial acceleration is removed. While commanded ON, the thrust target is bounded by minimum and maximum values:

$$T_{\text{cmd}} = \text{clamp}(m\|\mathbf{a}_{\text{req}}\|, 0.10T_{\text{max}}, T_{\text{max}}). \quad (14)$$

Actual thrust magnitude is rate-limited according to

$$\dot{T}_{\text{act}} = \text{clamp}(20(T_{\text{cmd}} - T_{\text{act}}), -20,000, +20,000) \text{ N s}^{-1}, \quad (15)$$

with endpoint checks. The actual thrust-direction state slews toward the desired direction with an illustrative maximum rate of  $2^\circ/\text{s}$ . This is a steering-response proxy, *not* a rigid-body attitude, torque, or gimbal simulation. Minimum ON throttle is likewise not a full multi-engine ignition/shutdown model.

### 4.5 Touchdown events, constraints, and optimization

The target site is represented by the unit vector  $\mathbf{s}$ . The spherical surface-arc landing error and touchdown speed are

$$e_{\text{site}} = R \cos^{-1} \left( \frac{\mathbf{r}}{\|\mathbf{r}\|} \cdot \mathbf{s} \right), \quad v_{\text{TD}} = \|\mathbf{v}(t_f)\|. \quad (16)$$

Integration terminates at surface contact,  $\|\mathbf{r}\| = R$ , or the dry-mass limit,  $m = m_{\text{dry}}$ . V17 tested 80 combinations of coast fraction and guidance time. A feasible solution required surface contact,  $v_{\text{TD}} \leq 2 \text{ m/s}$ ,  $e_{\text{site}} \leq 100 \text{ m}$ , and  $m_f \geq 125,000 \text{ kg}$ . Among feasible solutions meeting the more conservative  $v_{\text{TD}} \leq 1.85 \text{ m/s}$  target, the model selected the minimum-propellant case. This is a *discrete grid-search optimum*, not a proof of global optimality.

## 5 Why These Modeling Choices Were Made

- **Deorbit plus coast:** avoids treating the entire orbit-to-surface maneuver as one continuous powered burn.
- **Three-dimensional propagation:** measures both surface miss distance and terminal speed, rather than altitude alone.
- **Thrust and slew constraints:** prevents arbitrary instantaneous thrust-magnitude or direction changes in the guidance model.
- **Touchdown-speed margin:** avoids choosing the absolute minimum-fuel candidate at nearly the maximum permitted touchdown speed.
- **Convergence checks:** tests sensitivity of the reported results to numerical integration settings.
- **Perturbation trials:** explores sensitivity to assumed ignition-state and engine-performance uncertainty.

## 6 Numerical Checks and Limitations

**Optimization results:** 72 of 80 candidates met the basic feasibility constraints; 58 also met the 1.85 m/s preferred speed target. V17 selected coast fraction 0.928, guidance time 660 s, touchdown speed 1.8028 m/s, and landing error 3.6949 m.

**Solver convergence:** maximum integration steps of 2, 1, and 0.5 s gave identical displayed touchdown metrics. This supports numerical consistency at the reported precision but does not validate the underlying physical assumptions.

**Perturbation study:** all 60 of 60 illustrative trials met modeled landing constraints. The worst reported touchdown speed was 1.8054 m/s, worst miss distance was 3.696 m, and minimum margin above dry mass was 16,760.7 kg. Gaussian one-sigma perturbations were 10 m per position component, 0.1 m/s per velocity component, 100 kg ignition mass, 1% thrust scaling, and 0.5% specific-impulse scaling. These distributions are illustrative, so 60/60 successes should *not* be interpreted as a real mission reliability estimate.

**Physical limitations:** the model assumes a spherical, nonrotating Moon and a fixed inertial target; it excludes lunar gravity harmonics, terrain and hazards, navigation errors, full six-degree-of-freedom attitude dynamics, true gimbal and ignition logic, communications, structural loads, and verified Starship HLS performance data. V18 compared illustrative sites using orbital-plane accessibility proxies only; it did not compute site-specific descent propellant and is not needed for the present single-site budget.

## 7 MATLAB Files to Keep

| Priority       | File                                           | Purpose                                                         |
|----------------|------------------------------------------------|-----------------------------------------------------------------|
| <b>Primary</b> | <code>hls_integrated_descent_v17.m</code>      | Final trajectory, fuel optimization, convergence and robustness |
| Support        | <code>hls_integrated_descent_v16.m</code>      | Thrust-direction slew diagnostics                               |
| Support        | <code>hls_v15_validation.m</code>              | Earlier numerical-validation study; depends on V14 workspace    |
| Support        | <code>hls_integrated_descent_v14.m</code>      | Earlier engine-constraint baseline                              |
| Optional       | <code>hls_landing_site_comparison_v18.m</code> | Illustrative orbital-access screening, not landing fuel         |
| Archive        | V1–V13                                         | Development history; not final quantitative results             |

## 8 Recommended Figures and Captions

1. **Essential — V17 nominal descent:** altitude, speed, thrust and mass against powered-descent time. *Suggested caption:* “Simulated nominal lunar powered descent showing altitude, speed, engine thrust, and spacecraft mass over time.”
2. **Essential — V17 fuel versus touchdown speed:** candidate propellant consumption versus terminal speed, highlighting the margin-qualified selection. *Suggested caption:* “Tradeoff between powered-descent propellant consumption and touchdown speed across the guidance search.”
3. **Recommended — V17 robustness:** touchdown outcomes and remaining mass under 60 illustrative perturbations. *Suggested caption:* “Sensitivity of terminal performance and dry-mass margin to assumed initial-state and engine-performance variations.”
4. **Optional — V16 direction diagnostics:** commanded-versus-actual thrust direction and direction slew rate; useful only if discussing the steering approximation.

Do not use V18’s illustrative site chart to imply different powered-descent fuel costs. In MATLAB, export each figure at publication resolution with `exportgraphics(gcf, 'figure.png', 'Resolution', 300)` after selecting the relevant window; vector PDF is preferable for line plots when accepted by the report workflow.

## 9 Report-Ready Results Summary

A simplified three-dimensional lunar descent simulation was developed to estimate the propulsive velocity increment and propellant consumption required to land at  $88.811008^\circ$  S,  $123.690068^\circ$  E. Starting from an assumed 250,000 kg spacecraft in a 100 km circular lunar orbit, an idealized retrograde impulse of 19.46 m/s lowered the transfer perilune to 15 km and consumed approximately 1,413 kg of propellant. A coast phase was followed by feedback-guided powered descent with simplified thrust-magnitude, thrust-rate and thrust-direction constraints. Among 80 sampled guidance trajectories, the minimum-propellant solution satisfying a preferred touchdown-speed limit of 1.85 m/s consumed approximately 110,850 kg of powered-descent propellant and achieved 1.803 m/s terminal speed and 3.695 m landing error. The estimated powered-descent propulsive increment was 2,026.6 m/s, giving a combined orbit-to-touchdown budget of approximately 2,046.1 m/s and 112,263 kg propellant. These predictions depend on illustrative vehicle inputs and simplified lunar dynamics and are not verified Starship HLS flight requirements.